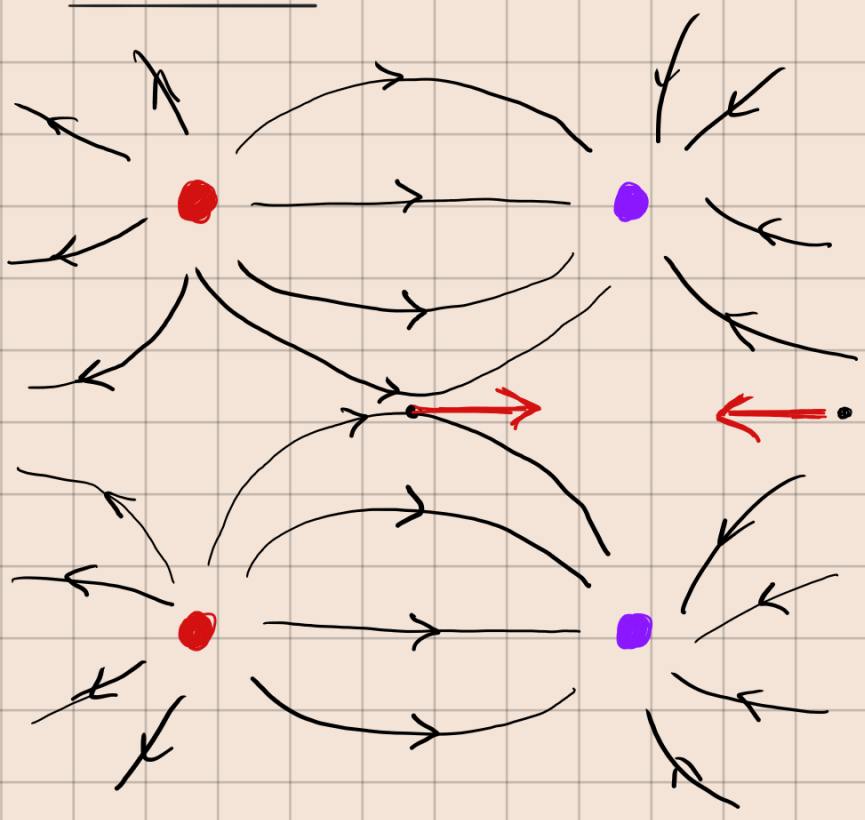
Organization:

Tuesday 10:15 – 11:45 (no break)

Come early (10:00) for questions.

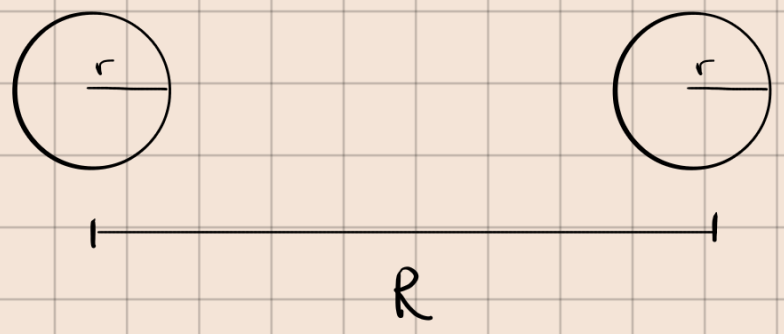
Ex 1



Lines going out of +,

into −.

Ex 2



a) Gravitational force: $F_G = G \dfrac{m^2}{R^2}$

with mass $m = V\rho = \frac{4}{3}\pi r^3 \rho$

$= 0.0475$ kg

$$\Rightarrow F_G = G\frac{m^2}{R^2} = 1.5 \times 10^{-13}\ \text{N}.$$

$$\left(\text{with } G = 6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)$$

b) Number of conduction electrons per sphere:

$$N = \frac{m \cdot N_A}{A}$$

N_A → Avogadro number

A → atomic mass

and total charge $Q = N \cdot e$.

After the transfer, one sphere has $+Q$, the other has $-Q$.

Coulomb force:

$$F_C = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q^2}{R^2} \overset{Q = Ne}{=} \frac{1}{4\pi\epsilon_0} \cdot \frac{N^2 e^2}{R^2}$$

$$\overset{N = \frac{m N_A}{A}}{=} \frac{1}{4\pi\epsilon_0} \cdot \frac{m^2 N_A^2 e^2}{A^2 R^2} \overset{m = \frac{4}{3}\pi r^3 \rho}{=} \frac{4\pi}{9\epsilon_0} \cdot \frac{r^6 \rho^2 N_A^2 e^2}{A^2 R^2}$$

$$= 4.38 \times 10^{18}\ \mathrm{N}$$

Compare to $F_G \approx 10^{-13}\ \mathrm{N}$!

c) When does F_G become equal to F_c?

$$F_c = F_G$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \cdot \frac{Q^2}{\cancel{R^2}} = G \frac{m^2}{\cancel{R^2}}$$

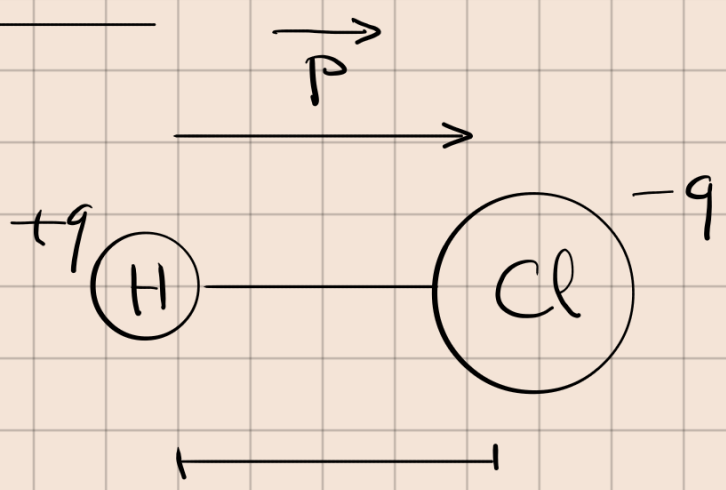
$$\Rightarrow \frac{Q}{m} = \sqrt{4\pi\epsilon_0 G}$$

$$= 8.6 \times 10^{-11} \, C/kg$$

Very small! For reference,

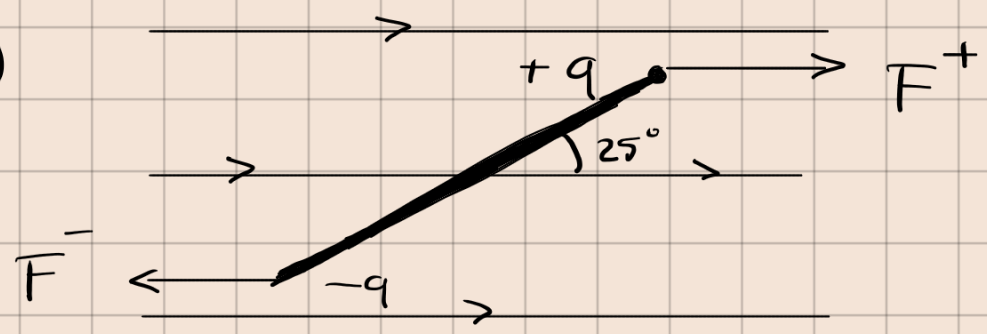
electron $\frac{e}{m} \approx 10^{11} \, C/kg$

Ex 3



a) $\vec{p} = q\vec{a} = 3.58 \times 10^{-30}\ C \cdot m$

b)



Forces: $F^+ = Eq$, $F^- = -Eq$

$\Rightarrow$ Torques: $M^{\pm} = \vec{r}^{\pm} \times \vec{F}^{\pm}$ (same direction) ↺ ↻

$= \frac{d}{2} Eq \sin\alpha$

Total torque: $M_T = dEq\sin\alpha$

$= 4.53 \times 10^{-27}\ N \cdot m$

The dipole will rotate and align with the E-field.

c) If the field is not homogenous,

$$E^+ \neq E^- \Rightarrow F^+ \neq F^-$$

So the dipole will move towards stronger field.

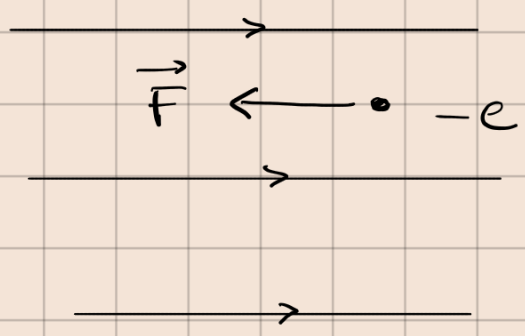
Ex 4

a) $$\frac{e}{m_e} = \frac{1.602 \times 10^{-19}\ C}{9.109 \times 10^{-31}\ kg} = 1.76 \times 10^{11}\ C/kg$$

b) $$\vec{F} = m\vec{a}$$

$$\Rightarrow |a| = \frac{|F|}{m_e} = |E|\frac{|e|}{m_e} = 1.76 \times 10^{13}\ m/s^2$$

Electron is accelerated in the direction opposite to E-field.



$$\vec{F} = -e\vec{E}$$